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Automating MARC Holdings Generation with AI: Lessons from the Field

Built with an AI coding assistant; runs without one.

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More detail, the figures and a PDF of this poster
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1 The problem, and the approaches I tried

The problem

Many libraries' serials holdings are stored only as human-readable, free-text **866** fields rather than as MARC-encoded **853/863** pairs. Converting by hand is slow, conventions vary, and hand-typed data is prone to typos and other errors.

> **The goal:** a tool that is portable across collections, works on messy holdings as they are (no manual clean-up), and never silently loses anything.

866 · as a person typed it

```
v. 27 no. 4-v. 31 no. 4
a  a  b  b  a  a  b  b

(April 1992-April 1996)
 j     i     j     i
```

↓

subfield	853 caption	863 held
$a volume	v.	27-31
$b issue	no.	4-4
$i year	(year)	1992-1996
$j month	(month)	04-04

```
853 31 $8 1 $a v. $b no. $i (year) $j (month)
863 40 $8 1.1 $a 27-31 $b 4-4 $i 1992-1996 $j 04-04
```

APPROACH 1

Write the script by hand

I adapted USC colleagues' script (Chung & Pendyala, 2023) over six months, teaching myself Python. The hardest part was one monolithic regular expression: weeks, if not months, in regex101 before it covered every pattern.

Pros: tuned to one collection; code I knew line by line.

Cons: only feasible on clean, consistent 866s, and not portable. Few institutions could write their own expression that way.

APPROACH 2

Ask a model to read the statements

Send the 866s to a large language model through its API, and ask it to pick out the volumes, issues and dates an 853/863 is built from.

Pros: the least coding knowledge; good results from minimal prompting.

Cons: tokens for every statement (free access stopped within two dozen records), and output that isn't guaranteed to be the same twice.

APPROACH 3 · CHOSEN

Use AI to build the tool

Describe in plain language what the tool should do, and have an AI coding assistant (Claude Code) write a deterministic Python application.

Pros: the same file converts the same way every time, with no model, API key or tokens when it runs.

Cons: a black box. I can confirm it works, and tests stop it regressing, without fully understanding why it works.

	1 · BY HAND	2 · A MODEL READS	3 · AI BUILDS THE TOOL
Coding knowledge	The most	The least	Some
Cost to run	Nothing	Tokens for every statement	Nothing
Same output every time	Yes	Not guaranteed	Yes
Other collections	Poorly	Not tested far enough	Designed for it

2 The approach chosen: AI-assisted development of a deterministic tool

How the Workbench reads a statement

1 · Cut into labelled pieces

```
v.  27  no.  4   -   v.  31  no.  4
VOL  #  ISS  #   –  VOL  #  ISS  #

(  April  1992  -  April  1996  )
     CHRON  YEAR  -  CHRON  YEAR
```

2 · Its shape

```
VOL ISS – VOL ISS(CHRON YEAR-CHRON YEAR)
```

```
v. <start_vol> no. <start_iss>-v. <end_vol> no. <end_iss>
(<start_month> <start_year>-<end_month> <end_year>)
```

1. Statements with the same shape are one group, whatever their numbers.
2. Each group gets a matching rule with a named slot for each value.
3. A cataloguer confirms what each value means, once per group.
4. The parser reads every statement; a pattern fills only what it cannot know.

What the cataloguer controls

- **On a pattern:** what each value is, its caption and level; skip a shape; test or edit the rule; reuse the library on another collection.
- **On one record:** edit an 866 before it is read, set this record's 853 (frequency, issues per volume, missing captions), skip it, leave a note.
- **For the whole file:** holdings level, subfield convention, strict reading, what to do with a value MARC has no place for, and whether to keep each 866.

Every decision is listed in a downloadable log.

The working rules

1. **Measure before you change.**
2. **Nothing disappears silently.**
3. **One reader of holdings.**
4. **If a fix needs the tests rewritten, stop.**
5. **Write the changelog for cataloguers.**

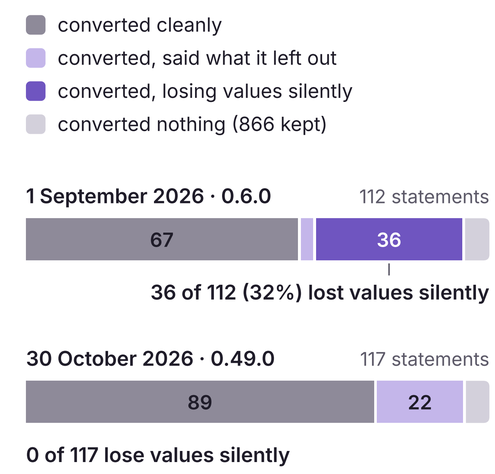
The human in the loop

The AI writes the code; the cataloguer decides what "correct" means. Two October bugs were found by the cataloguer, not the tests:

- `v. 58 Suppl. (Sep 2003)` lost "Suppl." when its pattern was confirmed (fixed in 0.32.0).
- `34 no 3, 4 (Summer, Autumn 1990)` ignored its confirmed caption (fixed in 0.34.0).

The turning point: silent losses

A silent loss writes the fields, removes the 866, and drops a volume or a month with nothing on screen to say so.

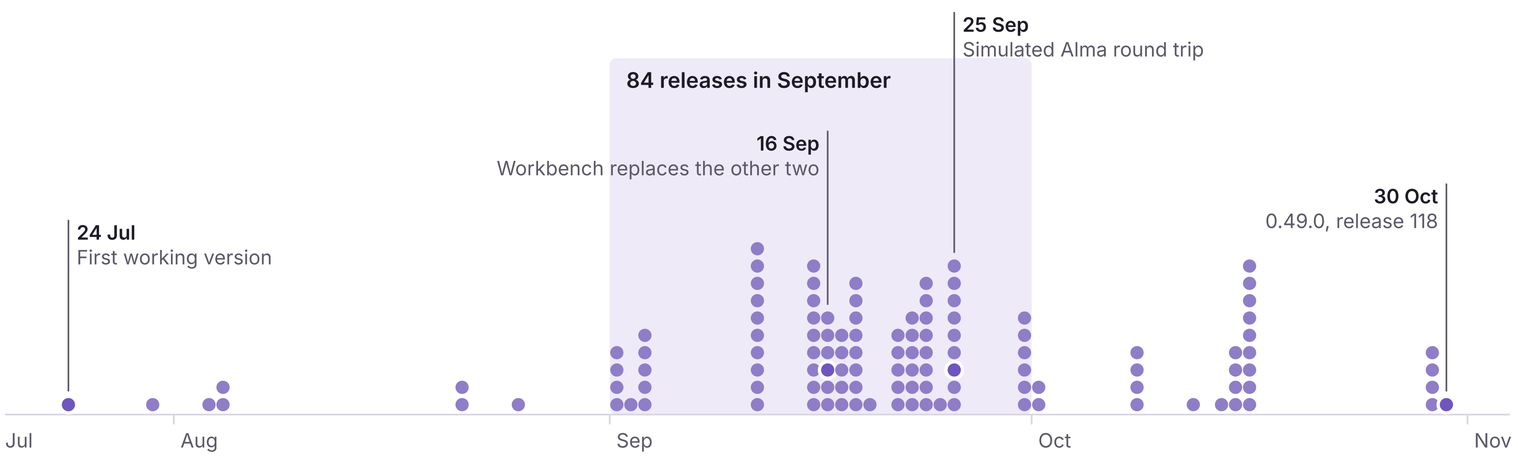


On real data

A real Alma export of 372 records:

1,057 holdings statements	**0** values unaccounted for
0 drift, simulated Alma round trip	**48/48** of Alma's own 866s reproduced

How it was built: 118 numbered versions in 14 weeks, most of them small



Each dot is one version. The Pattern Detector and Converter began as one-shot prompts; the Workbench combined them, and on 16 September replaced them.

Lessons, and what's next

- **A script reads what it was written for, and passes over the rest.** The Workbench makes the check part of the tool: every value is encoded, named as left out, or held with its 866 kept.
- **Decide whether you can work with code you don't fully understand.** A test suite and automated checks are what made it acceptable.
- **AI is at its best building the tool, not being it.**
- **Next:** testing on more real .mrc files from technical services colleagues, working toward a 1.0 release.

Try it: tools.matthewcodey.com/workbench · **Source:** github.com/bdmcodey/holdings-workbench

Chung, M., & Pendyala, P. C. (2023). Enhancing serials holdings data: A Pymarc-powered clean-up project. *Code4Lib Journal*, 58.